$HgCl_2$-Catalyzed monobenzylation and monoallylation reactions of 1-phenyl-1-alkynes

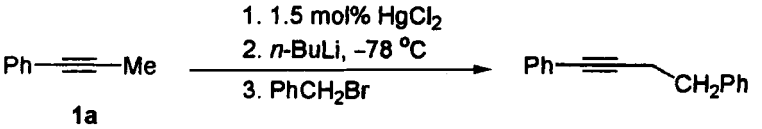
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Abstract A general procedure for the $HgCl_2$-mediated monolithiation reaction of 1-phenyl-1-alkynes was developed.

Keywords $HgCl_2$, monolithiation, 1-phenyl-1-alkynes, monobenzylation, monoallylation.

Carbon-carbon bond formation via a reaction of an organometallic reagent with an alkyl halide is one of the most efficient pathways in organic synthesis.[1] The reaction of organolithiums with organic halides is one of them.[1,2] Usually, organolithiums can be prepared via the following reactions: (1) the oxidative addition reaction of organic halides with the metallic lithium;[2] (2) halogen-lithium exchange reaction of organic halides;[3] (3) deprotonation reaction of compounds bearing activated C-H bond with BuLi, or LDA, *etc*.[4] Recently, during our study on the new methodology for the concise synthesis of 1,4-diaryl-1-alkynes we developed a new $HgCl_2$-catalyzed monolithiation reaction of 1-aryl-1-propynes.[5] The organolithiums formed via the reaction of 1-aryl-1-propynes with 1 equiv. of *n*-BuLi reacted smoothly with benzylic halides to afford 1,4-diaryl-1-propynes as the major products (Eq. 1).[5] In this paper, we wish to discuss our recent results on the monolithiation reaction of 1-phenyl-1-alkynes other than 1-aryl-1-propynes and the subsequent alkylation reaction (including benzylation and allylation), as well as the monoallylation reaction of 1-phenyl-1-propyne.



(1)

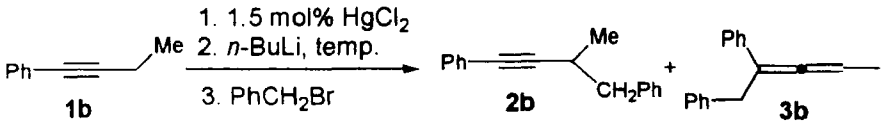
Published studies on lithiation of 1-phenyl-1-alkynes via their treatment with *n*-BuLi in ether revealed that bislithiation reaction immediately followed the monolithiation, and the bislithiation reaction was even faster than the monolithiation, thus, affording a mixture of a divalent propargylic anion and a

Received February 9, 1999; accepted March 8, 1999.

Project supported by the National Natural Science Foundation of China (Nos. 29525202 and 29872044) and the Shanghai Municipal Committee of Science and Technology (No. 98QA14001).

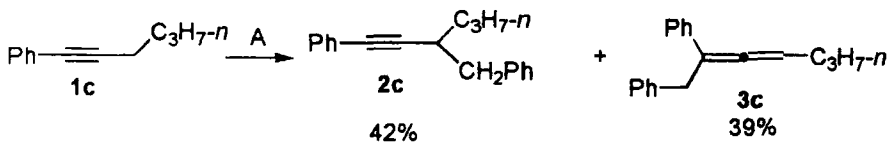
monovalent propargylic anion, which makes this reaction synthetically useless.[6] However, in THF the $HgCl_2$-mediated lithiation reaction of 1-aryl-1-propynes at $-78℃$ went smoothly and selectively, the corresponding bislithiation reaction was to the extent of $< 2\%$.[5] However, under the same reaction conditions, 1-phenyl-1-butyne did not react with *n*-BuLi in THF at $-78℃$, as shown by the subsequent reaction at $-78℃$ with $PhCH_2Br$. Luckily we found that the monolithiation could occur smoothly at room temperature. The subsequent reaction with $PhCH_2Br$ afforded 1,4-diphenyl-3-methyl-1-butyne (**2b**) in 67% yield and 1,2-diphenyl-2,3-pentadiene (**3b**) in 22% yield (Scheme 1).

Scheme 1



temp. (°C)	reaction time (h)	yield of **2b** (%)	yield of **3b** (%)
−78	10	0	0
25	5	67	22

The monolithiation of 1-phenyl-1-hexyne and the subsequent benzylation also went similarly (Eq. 2).



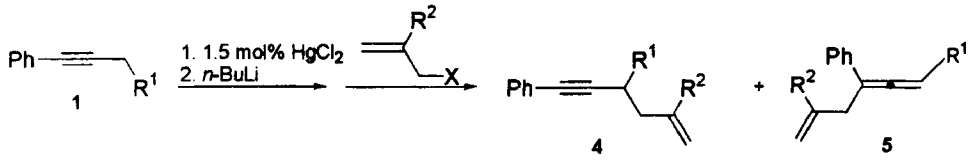
(2)

A: 1) 1.5 mol% $HgCl_2$; 2) *n*-BuLi, 25 °C; 3) $PhCH_2Br$

The propargylic lithiums formed by the $HgCl_2$-catalyzed monolithiation also reacted with allylic halides, some representative results are summarized in Table 1. The following points should be noted: (1) the monoallylation reaction of 1-phenyl-1-propyne afforded 1-phenyl-5-en-1-ynes as the major products (Entries 1—4, Table 1); (2) the monoallylation reaction of 1-phenyl-1-butyne (**1b**), 1-phenyl-1-hexyne (**1c**), and 1,4-diphenyl-1-butyne (**1d**) afforded the corresponding allenes as the major products (Entries 5—9, Table 1), probably due to the increase of the steric hindrance at the progarylic positions; (3) the treatment of 1,4-diphenyl-1-butyne with *n*-BuLi afforded the monolithiation product at 3-position highly chemoselectively, while the monolithiation at the benzylic 4-position[7] was not observed (Entry 9, Table 1 and Scheme 2); (4) in the absence of $HgCl_2$, the monoallylation of 1-phenyl-1-alkynes with *n*-BuLi in THF afforded the corresponding products in very low yields (Scheme 3). The similar reaction of 1-phenyl-1-propyne (**1a**) in ether did not occur, **1a** was recovered in 92% NMR yield.

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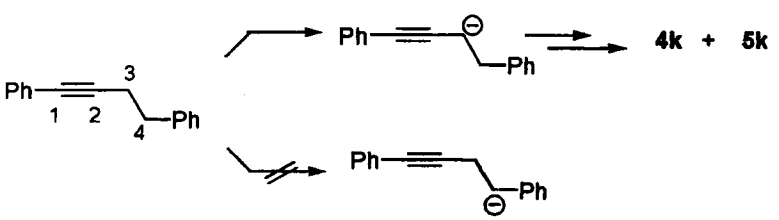
Table 1 Monoallylation reaction of 1-phenyl-1-alkynes[a]



Entry	Alkyne **1**	Allyl R^2	Halide X	Time (h)	Products[b]	(Yields %)[c]
1	R^1 = H (**1a**)	H	Br	6	**4a** (70)	**5a** (27)
2	**1a**	Cl	Cl	5	**4b** (60)	**5b** (37)
3	**1a**	Me	Cl	6	**4c** (48)	**5c** (45)
4	**1a**	Br	Br	4	**4d** (52)	**5d** (31)
5	R^1 = Me (**1b**)	H	Br	5	**4e** (25)	**5e** (51)
6	**1b**	Me	Cl	9	**4f** (36)	**5f** (50)
7	R^1 = *n*-Pr (**1c**)	H	Br	5	**4g** (30)	**5g** (53)
8	**1c**	Me	Cl	9	**4h** (32)	**5h** (48)
9	R^1 = Bn (**1d**)	H	Br	5	**4i** (28)	**5i** (48)

[a] The monolithiation reaction of **1a** was carried out at −78℃, while monolithiation reaction of **1b—d** was carried out at r.t. [b] Satisfactory ^{1}H NMR, IR, MS, and elemental analysis or HRMS data were obtained for all the new compounds. [c] Isolated yield.

Scheme 2

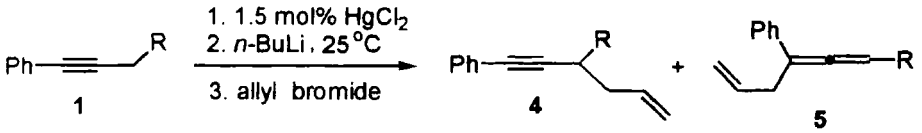


The reactivity of propargylic mercurys was reported to be sluggish, and their reactions with alkyl halides were slow and should be catalyzed by Lewis acid or $h\nu$.[8] Thus, as a control experiment, 1-phenyl-1-propargyl mercury bromide was synthesized according to a reported procedure.[9] Its reaction with allyl bromide was tried. No cross-coupling products **4a** and **5a** were detected and 1-phenyl-1-propargyl mercury bromide was recovered in 90% yield (Scheme 4), indicating that the monobenzylation and monoallylation of 1-phenyl-1-alkynes did not proceed via the reaction of organic halides with propargylic mercury halides, which, in principle, could be formed via the transmetalation reaction of the propargylic lithiums with $HgCl_2$. Thus, both the solvent effect of THF and the catalytic amount of $HgCl_2$ might stop the bislithiation reaction of 1-phenyl-1-alkynes.

In summary, we have developed a general $HgCl_2$-catalyzed monolithiation reaction of 1-phenyl-1-alkynes. The subsequent reaction with benzyl bromide and allylic halides afforded important synthetic intermediates, *i.e.*, alkynes and allenes, in high yields. The investigation of the reaction scope and the exact role of $HgCl_2$ are currently under way in our laboratory.

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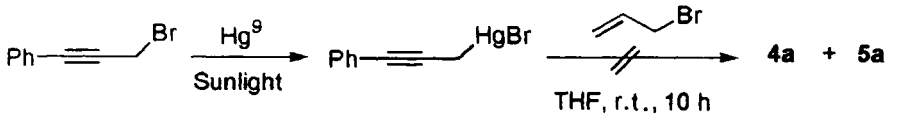
Scheme 3



1	reaction time (h)	4	5
1c (R = C_3H_7)	5	0	0
1b (R = CH_3)	5	0	0
1a (R = H)[a]	6	11	13

[a] The monolithiation reaction was carried out at −78 °C.

Scheme 4



Experimental

Materials

n-BuLi (from Aldrich, 1.6 mol/L in hexane), phenylacetylene (from Fluka or from Aldrich), allyl bromide (from Aldrich), and 3-chloro-2-methyl-1-propene (from TCI) are commercially available and used as it is. 1-Phenyl-1-propyne and 1-phenyl-1-butyne were prepared according to a published procedure[5,10] in 90% and 81%, respectively. After the addition of MeI (or EtI) and HMPA, the reaction was carried out at room temperature. The ^{1}H NMR data were the same as reported in References 11 and 12. 2,3-Dichloro-1-propene and 2,3-dibromo-1-propene were prepared according to a published procedure[13] in 70% and 49% yields, respectively. The ^{1}H NMR data were the same as reported in References 14 and 15. 3-Phenyl-3-propargyl mecury bromide was prepared from the sunlight-mediated reaction of propargyl bromide with mercury via a published procedure.[9] The NMR spectra were measured in $CDCl_3$ using TMS as the internal standard.

Monoallyation reaction of 1-phenyl-1-propyne

(a) *Synthesis of 1-phenyl-5-hexen-1-yne* (**4a**) *and 3-phenyl-1,2,5-hexatriene* (**5a**) *Typical procedure* To a solution of 1-phenyl-1-propyne (116 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%) in THF (5 mL) was added *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) at −78 ℃. After 10 min, allyl bromide (145 mg, 1.2 mmol) was added at −78 ℃ and the reaction was allowed to warm up to room temperature naturally. After 6 h at room temperature, the reaction was quenched with saturated NH_4Cl. Extracting with petroleum ether (60—90 ℃), washing with saturated NaCl, drying over $MgSO_4$, and chromatography on silica gel (petroleum ether) afforded 109 mg (70%) of **4a** and 42 mg (27%) of

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5a. **4a**: liquid. ν_{max}(film): 2235 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 2.22(t, J = 6.67 Hz, 2H), 2.54(t, J = 6.67 Hz, 2H), 4.88—5.12(m, 2H), 5.23—5.42(m, 1H), 7.02—7.40(m, 5H). m/z(EI): 156(M^+, 32.78), 115(100). Anal. $C_{12}H_{12}$. Calcd.: C, 92.26; H, 7.74. Found: C, 92.06; H, 7.95. **5a**: liquid. ν_{max}(film): 1938 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 2.98—3.40(m, 2H), 4.87—5.34 (m, 4H), 5.86—6.07(m, 1H), 7.08—7.51(m, 5H). m/z(EI): 156(M^+, 37.44), 115(100); HRMS 156.0900 (M^+, $C_{12}H_{12}$ requires 156.0939).

The following compounds were prepared similarly.

(b) 1-*Phenyl*-5-*chloro*-5-*hexen*-1-*hexyne* (**4b**) *and* 3-*phenyl*-5-*chloro*-1,2,5-*hexatriene* (**5b**) Using 1-phenyl-1-propyne (116 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and 2,3-dichloro-1-propene (1.2 mmol, 133 mg) gave 114 mg (60%) of **4b** and 70 mg (37%) of **5b**. **4b**: liquid. ν_{max} (film): 2200 cm^{-1}. δ_H (300 MHz, $CDCl_3$): 2.55—2.90(m, 4H), 5.25—5.40(m, 2H), 7.12—7.51(m, 5H). m/z(EI): 192(M^+ (^{37}Cl), 0.21), 190(M^+ (^{35}Cl), 0.58), 155(100). Anal. $C_{12}H_{11}Cl$. Calcd.: C, 75.59; H, 5.81. Found: C, 75.42; H, 5.95. **5b**: liquid. ν_{max}(film): 1937 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 3.39—3.48(m, 2H), 5.02—5.12(m, 2H), 5.15—5.21(m, 2H), 7.14—7.58(m, 5H). m/z(EI): 190(M^+, 10.15), 155(100). Anal. $C_{12}H_{11}Cl$. Calcd.: C, 75.59; H, 5.81. Found: C, 75.03; H, 5.86.

(c) 5-*Methyl*-1-*phenyl*-5-*hexen*-1-*yne* (**4c**) and 5-*methyl*-3-*phenyl*-1,2,5-*hexatriene* (**5c**) Using 1-phenyl-1-propyne (116 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and 3-chloro-2-methyl-1-propene (1.2 mmol, 107 mg) gave 82 mg (48—60%) of **4c** and 77 mg (37—45%) of **5c**. **4c**: liquid. ν_{max}(film): 2200 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 1.80(s, 3H), 2.35(t, J = 7.47 Hz, 2H), 2.56(t, J = 7.47 Hz, 2H), 4.82(s, 2H), 7.20—7.50(m, 5H). m/z EI): 170(M^+, 8.08), 115(100). Anal. $C_{13}H_{14}$. Calcd: C, 91.71; H, 8.29. Found: C, 91.07; H, 8.62. **5c**: liquid. ν_{max}(film): 1940 cm^{-1}, δ_H(300 MHz, $CDCl_3$): 1.88 (s, 3H), 3.28(s, 2H), 4.91(t, 2H, J = 0.94 Hz), 5.12(s, 2H), 7.21—7.56(m, 5H). m/z (EI): 170(M^+, 23.58), 155(100); HRMS 170.1104 (M^+, $C_{13}H_{14}$ requires 170.1096).

(d) 5-*Bromo*-1-*phenyl*-5-*hexen*-1-*yne* (**4d**) *and* 5-*bromo*-3-*phenyl*-1,2,5-*hexatriene* (**5d**) Using 1-phenyl-1-propyne (116 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and 2,3-dibromo-1-propene (1.2 mmol, 240 mg) to give 122 mg (52%) of **4d** and 73 mg (31%) of **5d**. **4d**: liquid. ν_{max} (film): 2200 cm^{-1}. δ_H (300 MHz, $CDCl_3$): 2.58—2.90(m, 4H), 5.53(d, J = 1.55 Hz, 1H), 5.72(d, J = 1.55 Hz, 1H), 7.16—7.52(m, 5H). m/z(EI): 236(M^+ (^{81}Br), 6.65), 234(M^+ (^{79}Br), 7.27), 153(100). HRMS 234.9942, 232.9976 ($M^+ - 1$, $C_{12}H_{10}Br$ requires (^{81}Br) 234.9946, (^{79}Br) 232.9966). **5d**: liquid. ν_{max}(film): 1940 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 3.63(s, 2H), 5.17(t, 2H, J = 2.09 Hz), 5.54(s, 1H), 5.73(s, 1H), 7.08—7.55(m, 5H). m/z(EI): 236(M^+ (^{81}Br), 5.13), 234(M^+ (^{79}Br), 5.33), 155 (100). Anal. $C_{12}H_{11}Br$. Calcd.: C, 61.30; H, 4.72. Found C, 61.69; H, 5.41.

Monobenzylation reaction of 1-*phenyl*-1-*alkynes*

(a) *Synthesis of* 3-*methyl*-1,4-*diphenyl*-1-*butyne* (**2b**) *and* 1,2-*diphenyl*-2,3-*pentadiene* (**3b**)

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Typical procedure To a solution of 1-phenyl-1-butyne (130 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%) in THF (5 mL) was added *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) at −78℃. Then the reaction was warmed up to room temperature immediately. After 0.5 h at room temperature, benzyl bromide (205 mg, 1.2 mmol) was added. Then the reaction was quenched with saturated NH_4Cl after 6 h. Extracting with petroleum ether (60—90℃), washing with saturated NaCl, drying over $MgSO_4$, and chromatography on silica gel (petroleum ether) afforded 138 mg (67%) of **2b** and 45 mg (22%) of **3b**. **2b**: liquid. ν_{max}(film): 2200 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 1.18(d, J = 6.40 Hz, 3H), 2.62—2.75(m, 1H,), 2.83(d, J = 7.95 Hz, 3H), 7.02—7.30(m, 10H). δ_C: 20.633, 28.643, 43.245, 81.683, 94.137, 124.018, 125.900, 126.344, 127.580, 128.201, 129.406, 131.564, 139.658. *m/z*(EI): 220(M^+, 25.12), 58(100). Anal. $C_{17}H_{16}$. Calcd.: C, 92.68; H, 7.32. Found: C, 92.50; H, 7.39. **3b**: liquid. ν_{max}(film): 1720 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 1.72 (d, 3H, J = 7.06 Hz), 3.70—3.95(m, 2H), 5.35—5.47(m, 1H), 7.10—7.50(m, 10H). *m/z* (EI): 220(M^+, 41.28), 205(100); HRMS 220.1239 (M^+, $C_{17}H_{16}$ Calcd.: 220.1248).

The following compounds were prepared similarly.

(b) *1-Phenyl-3-benzyl-1-hexyne* (**2c**) *and 1,2-diphenyl-2,3-heptadiene* (**3c**) Using 1-phenyl-1-hexyne (158 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (from Aldrich, 1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and benzyl bromide (205 mg, 1.2 mmol) afforded 104 mg (42%) of **2c** and 97 mg (39%) of **3c**. **2c**: liquid. ν_{max}(film): 2220 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 0.90(t, J = 7.30 Hz, 3H), 1.38—1.80(m, 4H), 2.65—2.95(m, 3H), 6.75—7.50(m, 10H). *m/z*(EI): 248(M^+, 1.43), 105(100). HRMS 248.1579 (M^+, $C_{19}H_{20}$ Calcd.: 248.1565). **3c**: ν_{max}(film): 1920 cm^{-1}. δ_H(300 MHZ, $CDCl_3$): 0.98 (t, J = 7.30 Hz, 3H), 1.44—1.58(m, 2H), 2.04—2.18 (m, 2H), 3.88(d, J = 1.55 Hz, 2H), 5.48—5.58(m, 1H), 7.18—7.62(m, 10H). *m/z*(EI): 248(M^+, 5.13), 115(100). HRMS 248.1573 (M^+, $C_{19}H_{20}$ Calcd.: 248.1565).

Monoallylation reaction of 1-phenyl-1-alkynes

(a) *Synthesis of 3-methyl-1-phenyl-5-ene-1-hexyne* (**4e**) *and 3-phenyl-1,3,5-heptatriene* (**5e**) Using the procedure described for monobenzylaion, except that an allylic halide was used instead of benzyl bromide, starting from 1-phenyl-1-butyne (130 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and allyl bromide (145 mg, 1.2 mmol), the reaction afforded 43 mg (25%) of **4e** and 88 mg (51%) of **5e**. **4e**: liquid. ν_{max}(film) 2233 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 1.25(d, J = 6.71 Hz, 3H), 2.22—2.45(m, 2H), 2.65—2.83(m, 1H), 5.08 dd, J_1 = 9.32 Hz, J_2 = 1.88 Hz, 1H), 5.15(dd, J_1 = 16.07 Hz, J_2 = 1.88 Hz, 1H), 5.88—6.10(m, 1H), 7.12—7.55(m, 5H). *m/z*(EI): 170(M^+, 10.69), 129(100). HRMS 170.1087 (M^+, $C_{13}H_{14}$ Calcd.: 170.1096). **5e**: liquid. ν_{max}(film): 1920 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 1.69 (d, J = 7.04 Hz, 3H), 3.08(d, J = 7.35 Hz, 2H), 4.98(d, J = 8.72 Hz, 1H), 5.09(d, J = 18.21 Hz, 1H), 5.32—5.45(m, 1H), 5.78—5.92(m, 1H), 7.02—7.48(m, 5H). *m/z*(EI): 170 (M^+, 12.69), 43(100). HRMS 170.1075(M^+, $C_{13}H_{14}$ Calcd.: 170.1096).

(b) *3,5-Dimethyl-1-phenyl-5-hexen-1-yne* (**4f**) *and 2-methyl-4-phenyl-1,4,5-heptatriene* (**5f**)

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Using 1-phenyl-1-butyne (130 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and 3-chloro-2-methyl-1-propene (107 mg, 1.2 mmol) afforded 66 mg (36%) of **4f** and 92 mg (50%) of **5f**. **4f**: liquid. ν_{max} (film): 2220 cm^{-1}. δ_H (300 MHz, $CDCl_3$): 1.28(d, J = 7.04 Hz, 3H), 1.82(s, 3H), 2.18(dd, J_1 = 7.35 Hz, J_2 = 7.11 Hz, 1H), 2.35(dd, J_1 = 7.35 Hz, J_2 = 7.11 Hz, 1H), 2.74(m, 1H), 4.76—4.90(m, 2H), 7.13—7.42(m, 5H). *m/z* (EI): 184(M^+, 6.34), 129(100). Anal. $C_{14}H_{16}$ Calcd.: C, 91.25; H, 8.75. Found: C, 90.58; H, 8.95. **5f**: liquid. ν_{max}(film): 1994 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 1.50—1.90(m, 6H), 3.09(s, 2H), 4.76(s, 2H), 5.35—5.46(m, 1H), 7.02—7.45(m, 5H). *m/z*(EI) 184(M^+, 14.22), 169(100). HRMS 184.1266 (M^+, $C_{14}H_{16}$ Calcd.: 184.1252).

(c) 3-(*n-Propyl*)-1-*phenyl*-5-*hexen*-1-*yne* (**4g**) *and* 4-*phenyl*-1,4,5-*nonatriene* (**5g**) Using 1-phenyl-1-butyne (158 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and allyl bromide (145 mg, 1.2 mmol) afforded 59 mg (30%) of **4g** and 105 mg (53%) of **5g**. **4g**: ν_{max} (film): 2210 cm^{-1}. δ_H (300 MHz, $CDCl_3$): 0.94(t, J = 7.04 Hz, 3H), 1.39—1.72(m, 4H), 2.33(t, J = 6.79 Hz, 2H), 2.56—2.72(m, 1H), 5.09(dd, J_1 = 9.26 Hz, 1H, J_2 = 1.86 Hz, 1H), 5.15(dd, J_1 = 18.21 Hz, J_2 = 1.86 Hz, 1H), 5.88—6.06(m, 1H), 7.21—7.46(m, 5H). *m/z*(EI): 199(M^+ + 1, 74.08), 91(100). Anal. $C_{15}H_{18}$ Calcd.: C, 90.85; H, 9.15%. Found: C, 90.50; H, 9.04. **5g**: liquid. ν_{max}(film): 1935 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 0.98(t, J = 7.36 Hz, 3H), 1.40—1.71(m, 2H), 2.13(q, J_1 = 7.19 Hz, 2H), 3.24(d, J = 6.85 Hz, 2H), 5.08(dd, J_1 = 10.08 Hz, J_2 = 1.49 Hz, 1H), 5.19(dd, J_1 = 18.00 Hz, J_2 = 1.49 Hz, 1H), 5.45—5.60(m, 1H), 5.86—6.10(m, 1H), 7.02—7.65(m, 5H). *m/z*(EI): 198(M^+, 2.64), 43(100); HRMS 198.1409(M^+, $C_{15}H_{18}$ Calcd.: 198.1409).

(d) 5-*Methyl*-3-(*n-propyl*)-1-*phenyl*-5-*hexen*-1-*yne* (**4h**) *and* 2-*methyl*-4-*phenyl*-1,4,5-*nonatriene* (**5h**) Using 1-phenyl-1-butyne (128 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and 3-chloro-2-methyl-1-propene (107 mg, 1.2 mmol) afforded 68 mg (32%) of **4h** and 102 mg (48%) of **5h**. **4h**: liquid. ν_{max}(film): 2231 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 0.91(t, J = 6.62 Hz, 3H), 1.25—1.62(m, 5H), 1.68—1.86(m, 2H), 2.12—2.38(m, 2H), 2.61—2.82(m, 1H), 4.61—4.90(m, 2H), 7.02—7.42(m, 5H); *m/z*(EI): 212(M^+, 0.52), 115(100); Anal. $C_{16}H_{20}$ Calcd.: C, 90.51; H, 9.49. Found: C, 90.72; H, 9.91. **5h**: liquid. ν_{max}(film): 1941 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 0.90(t, J = 7.33 Hz, 3H), 1.38—1.58(m, 2H), 1.73(s, 3H), 1.96—2.25(m, 2H), 3.11(s, 2H), 4.75(s, 2H), 5.43(t, J = 6.59 Hz, 1H), 7.05—7.48(m, 5H). *m/z*(EI): 212(M^+, 7.34), 115(100); HRMS 212.1554(M^+, $C_{16}H_{20}$ Calcd.: 212.1565).

(e) 1-*Phenyl*-3-*benzyl*-1-*hexyne* (**4i**) *and* 1,4-*diphenyl*-1,4,5-*heptatriene* (**5i**) Using 1,4-diphenyl-1-butyne (206 mg, 1.0 mmol), $HgCl_2$ (4 mg, 1.5 mol%), *n*-BuLi (1.6 mol/L in hexane, 0.63 mL, 1.0 mmol) and allyl bromide (145 mg, 1.2 mmol) afforded 69 mg (25%) of **4i** and 118 mg (48%) of **5i**. **4i**: liquid. ν_{max} (film): 2205 cm^{-1}. δ_H (300 MHz, $CDCl_3$): 2.26—2.45(m, 2H), 2.82—2.98(m, 3H), 5.08—6.22(m, 2H), 5.90—6.10(m, 1H), 7.15—7.50(m, 10H). *m/z* (EI): 246(M^+, 5.00), 84(100); HRMS 246.1421(M^+, $C_{19}H_{18}$ Calcd.: 246.1408). **5i**: liquid. ν_{max}(film): 1945 cm^{-1}. δ_H(300 MHz, $CDCl_3$): 3.15(d, J = 5.24 Hz, 2H), 3.45(d, J = 7.14 Hz,

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2H), 4.95—5.15(m, 2H), 5.58—5.69(m, 1H), 5.85—6.00(m, 1H), 7.05—7.52(m, 10H). *m/z*(EI): 246(M^+, 15.47), 205(100). HRMS 246.1421(M^+, $C_{19}H_{18}$ Calcd.: 246.1409).

Reaction of 3-phenyl-3-propargyl mercury bromide with allyl bromide

To a solution of propargyl mercury bromide (20 mg, 0.05 mmol) in THF (5 mL) was added allyl bromide (7.3 mg, 0.06 mmol) at −78℃. Then the reaction was allowed to warm up to room temperature naturally. After 10 h at room temperature, the reaction was quenched with saturated NH_4Cl. Extraction with petroleum ether (60—90℃), washing with saturated NaCl, drying over $MgSO_4$, filtering, and removal the solvents afforded the crude product, which was analyzed by ^1H NMR spectra. No cross-coupling-products **4a** or **5a** were detected. Propargyl mercury bromide was recovered in 90% yield as determined by ^1H NMR spectra using CH_2Br_2 as the internal standard.

References

1. Trost, B.M.; Fleming I., *Comprehensive Organic Synthesis*, Vol. 3, Pergamon Press, Oxford, 1991.
2. Schlosser, M., in "*Organometallics in Synthesis-A Manual*", Ed. Schlosser, M., John & Wiley, 1994, p. 1.
3. a) Merrill, R.E.; Negishi E., *J. Org. Chem.*, **39**, 3452(1974).
   b) Negishi, R.E.; Swanson, D.R.; Rousset C.,J., *J. Org. Chem.*, **55**, 5406(1990).
4. a) Corey, E.J.; Kirst, H.A., *Tetrahedron Lett.*, 5041(1968).
   b) Corey, E.J.; Christoph, R., *Tetrahedron Lett.*, **23**, 719(1982).
   c) Wang, K.K.; Nikam, S.S.; Ho, C.D., *J. Org. Chem.*, **48**, 5376(1983).
   d) Nikam, S.S.; Wang, K.K., *J. Org. Chem.*, **50**, 2193(1985).
5. Ma, S.; Wang, L., *J. Org. Chem.* **62**, 3497(1998).
6. a) De Jong, R.L.P.; Brandsma, L., *J. Organometal. Chem.*, **312**, 277(1986).
   b) De Jong, R.L.P.; Brandsma, L., *ibid.*, **238**, C17(1982).
   c) Becker, J.Y.; Klein, J.J., *J. Organometal. Chem.*, **157**, 1(1978).
   d) Becker, J.Y., *J. Organometal. Chem.*, **118**, 247(1976).
7. a) Eberhardt, G.G.; Butte, W.A., *J. Org. Chem.*, **29**, 2928(1964).
   b) Lochmann, L.; Petrnek, J., *Tetrahedron Lett.*, **32**, 1483(1991).
8. For a review, see: Larock, R.C., *Organomercury Compounds in Organic Synthesis*, Springer-Verlag, Berlin, 1985, p. 249; For $h\nu$-mediated reactions, see: a) Russell, G.A.; Guo, D.; Khanna, R.K., *J. Org. Chem.*, **50**, 3423(1985).
   b) Russell, G.A.; Ngoviwatchai, P.; Wu, Y.W., *J. Am. Chem. Soc.*, **111**, 4921(1989).
9. Larock, R.C.; Chow, M.-S., *Tetrahedron Lett.*, **25**, 2727(1984).
10. Corey, E.J.; Katzenellenbogen, J.A.; Gilman, N.W.; Roman, S.A.; Erickson, B.W., *J. Am. Chem. Soc.* **90**, 5618(1968).
11. Ponchert, C.J., *Aldrich Library of NMR Spectra*, Aldrich Chemical Company, 2nd ed., 2, 961C.
12. Asahi Research Center Co., *Handbook of Proton-NMR spectra and Data*, Vol. 7, 5341, Academic Press, Tokyo, 1986.
13. Lespieau, R.; Bourquel, R., *Org. Synth.* Coll. Vol. **1**, 203(1932).
14. Ponchert, C.J., *Aldrich Library of NMR Spectra*, Aldrich Chemical Company, 2nd ed., 1, 90D.
15. Ponchert, C. J., *Aldrich Library of NMR Spectra*, Aldrich Chemical Company, 2nd ed., 1, 91A.

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